MATH1101: Differential Calculus & Analytical Geometry

Complete Exam Notes · 1st Year 1st Semester · PUST · Module 1 (Units 1, 2 & 3): Functions, Graphs, Limits & Continuity

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Course & Exam Overview

Course Code & Title: MATH1101 — Differential Calculus and Analytical Geometry (Credit: 3.0).

1. **Part A: Differential Calculus** (Modules 1–4, Units 1–13) — Contributes **4 questions** out of 7 (or 5 out of 8).
2. **Part B: Analytical Geometry (2D & 3D)** (Modules 5–8, Units 14–24) — Contributes **3 questions** out of 7.

Semester Exam Pattern: Total Marks = 70. Answer **5 questions out of 7** (14 marks per full question, typically divided into a, b, c parts of 4 to 5.5 marks each) in a single unified question paper.

Units Covered in Module 1:

- **Unit 1:** Concept of Function, Classification, Domain and Range.
- **Unit 2:** Graphs of Functions (Shifting, Scaling, Reflecting, Piecewise), Inverse, Even, Odd & Periodic Functions.
- **Unit 3:** Concept & Existence of Limit, ϵ-δ Formal Proof, Infinite Limits, Limits at Infinity, Continuity & Discontinuity.

Master Roadmap (MATH 1101)

Mod.	Units & Core Syllabus Topics	Main Sources	Status
1	Units 1–3: Functions, Domain/Range, Graphs, Inverse, Limits & Continuity	Class Notes · Anton Ch. 0, 1 · PUST 20–23	✓ **This file**
2	Units 4–5: Differentiability, Geometric Meaning, Extrema, y_n & Leibniz	Class Notes · Anton Ch. 2, 4 · PUST 20–23	Module 2
3	Units 6–9: Rolle's & MVT, Taylor/Maclaurin, L'Hôpital & Curve Tracing	Class Notes · Anton Ch. 3, 4, 9 · Handout	Module 3
4	Units 10–13: Tangents, Asymptotes, Curvature, Partial Deriv., Euler & Jacobian	Akib Notes · Anton Ch. 13 · PUST 20–23	Module 4
5	Units 14–15: 2D Coordinates, Transformation of Axes & Pair of Straight Lines	Anton Ch. 10 · Rahman & Bhattacharjee	Module 5
6	Units 16–18: Conic Reduction, Parabola, Ellipse, Hyperbola & System of Circles	Anton Ch. 10 · Rahman & Bhattacharjee	Module 6
7	Units 19–21: 3D Coordinates, DCs/DRs, Projections, Planes & Skew Lines (S.D.)	Anton Ch. 11 · Rahman & Bhattacharjee	Module 7
8	Units 22–24: Sphere, Cone, Right Circular Cylinder & 3D Central Conicoids	Anton Ch. 11 · Rahman & Bhattacharjee	Module 8

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1 Unit 1: Functions, Classification, Domain and Range

1.1 Definition and Ways of Expressing a Function

Definition (Function): If two variables x and y are related such that for every permissible input value of x in a set D, there corresponds **one and only one** finite output value of y, then y is called a function of x, written as:

$$y = f(x)$$

where x is the **independent variable** (input) and y is the **dependent variable** (output).

- **Valid Function Example:** $y = f(x) = x^2 + 1$.

  For $x = 1$, $\quad f(1) = 1^2 + 1 = 2$ [Unique output for $x = 1$]
  For $x = 2$, $\quad f(2) = 2^2 + 1 = 5$ [Unique output for $x = 2$]

- **Non-Function Counterexample:** $y^2 = x \implies y = \pm\sqrt{x}$. For a single input $x = 4$, it produces two distinct outputs $y = +2$ and $y = -2$, so $y^2 = x$ is not a function of x.

Four Ways of Expressing a Function:

1. **Analytically / Algebraically (by formula):** e.g., $y = f(x) = x^2 + 1$.
2. **Numerically (by table of values):** Listing input-output pairs (x_i, y_i).
3. **Geometrically (by graph):** Plotting all points $(x, f(x))$ in the xy-plane.
4. **Verbally (in words):** Describing the correspondence in plain language.

1.2 Classification of Functions

Functions in Differential Calculus are classified into two main families: **Algebraic Functions** and **Transcendental Functions**.

1.3 Domain and Range of Functions

- **Definition (Domain, D_f):** The set of all real input values $x \in \mathbb{R}$ for which $f(x)$ yields a real, finite value.
- **Definition (Range, R_f):** The set of all output values $y = f(x)$ obtained as x varies throughout D_f.

Main Family	Sub-Type	General Form & Standard Examples
1. Algebraic	(a) Polynomial	$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ $(n \in \mathbb{N} \cup \{0\}, a_n \neq 0)$. Domain is $\mathbb{R}$.
	(b) Rational	$f(x) = \dfrac{P(x)}{Q(x)}$, where $P(x), Q(x)$ are polynomials and $Q(x) \neq 0$.
	(c) Irrational	Expressions involving radicals or fractional powers, e.g., $f(x) = \sqrt{x^2 - 5x + 6}$.
	(d) Piecewise	Defined by different formulas on different parts of the domain, e.g., $f(x) = \lvert x \rvert$.
2. Transcendental	(a) Trigonometric	$\sin x, \cos x, \tan x, \cot x, \sec x, \csc x$ (angle x measured in radians).
	(b) Inverse Trig	$\sin^{-1} x, \cos^{-1} x, \tan^{-1} x, \cot^{-1} x, \sec^{-1} x, \csc^{-1} x$ (on principal branches).
	(c) Exponential	$f(x) = a^x$ $(a > 0, a \neq 1)$, especially $f(x) = e^x$.
	(d) Logarithmic	$f(x) = \log_a x$ $(a > 0, a \neq 1, x > 0)$, especially $\ln x = \log_e x$.

Important Note: Core Conditions for Domain Restrictions

1. **Rational Function** $\dfrac{P(x)}{Q(x)}$**:** Defined when Denominator $Q(x) \neq 0$.
2. **Even Root in Numerator** $\sqrt{g(x)}$**:** Defined when $g(x) \geq 0$.
3. **Even Root in Denominator** $\dfrac{1}{\sqrt{g(x)}}$**:** Defined when $g(x) > 0$ (strict inequality since denominator $\neq 0$).
4. **Logarithmic Function** $\ln(g(x))$**:** Defined when $g(x) > 0$.
5. **Inverse Sine/Cosine** $\sin^{-1}(g(x))$ **or** $\cos^{-1}(g(x))$**:** Defined when $-1 \leq g(x) \leq 1$.

1.4 Solved Problems on Domain and Range (Written in Exam Style)

Problem 1.1 (Class Lecture): Find the domain and range of $f(x) = \dfrac{1}{x-3}$**.**

Solution (Exam Style):

Step 1 (Domain): Given,

$$f(x) = \frac{1}{x-3}$$

Here, $f(x)$ takes real and finite values for all $x \in \mathbb{R}$ except when the denominator is zero:

$$x - 3 = 0$$
$$\implies x = 3 \quad \text{[Excluded point where } f(x) \text{ is undefined]}$$

$\therefore$ Domain, $D_f = \mathbb{R} \setminus \{3\}$ or $(-\infty, 3) \cup (3, \infty)$.

Step 2 (Range): Let,

$$y = \frac{1}{x-3}$$
$$\implies x - 3 = \frac{1}{y}$$
$$\implies x = \frac{1}{y} + 3$$
$$\implies x = \frac{1+3y}{y} \quad \text{[Undefined when } y = 0\text{]}$$

Here, x is real and finite for all $y \in \mathbb{R}$ except $y = 0$.

$\therefore$ Range, $R_f = \mathbb{R} \setminus \{0\}$ or $(-\infty, 0) \cup (0, \infty)$.

Problem 1.2 (Class Lecture): Find the domain and range of $f(x) = \sqrt{x^2 - 5x + 6}$.

Solution (Exam Style):

Step 1 (Domain): Given,

$$f(x) = \sqrt{x^2 - 5x + 6} = \sqrt{(x-2)(x-3)}$$

For $f(x)$ to be real, the expression inside the square root must be non-negative:

$$x^2 - 5x + 6 \geq 0$$
$$\implies (x-2)(x-3) \geq 0 \quad \text{[Critical boundary points: } x = 2,\ 3\text{]}$$

Dividing the real line at $x = 2$ and $x = 3$ into three intervals:

Interval	**Sign of** $(x-2)$	**Sign of** $(x-3)$	**Sign of** $(x-2)(x-3)$	**Status**
$(-\infty, 2)$	$-$	$-$	$(-) \times (-) = +$	Valid (> 0)
$(2, 3)$	$+$	$-$	$(+) \times (-) = -$	Invalid (< 0)
$(3, \infty)$	$+$	$+$	$(+) \times (+) = +$	Valid (> 0)

Checking the boundary points:

At $x = 2$, $\quad f(2) = \sqrt{(2-2)(2-3)} = 0$ [Defined, so $x = 2$ is included]
At $x = 3$, $\quad f(3) = \sqrt{(3-2)(3-3)} = 0$ [Defined, so $x = 3$ is included]

$\therefore$ Domain, $D_f = (-\infty, 2] \cup [3, \infty)$.

Step 2 (Range): Since $f(x) = \sqrt{x^2 - 5x + 6}$ is a principal square root, $f(x) \geq 0$ for all $x \in D_f$. At $x = 2$ or 3, $f(x) = 0$, and as $x \to \pm\infty$, $f(x) \to +\infty$ continuously.

$\therefore$ Range, $R_f = [0, \infty)$.

Problem 1.3 (Class Lecture & PUST EECE 2020–21 Exam Q1a [5 Marks]): Find the domain and range of $f(x) = \dfrac{x-2}{3-x}$.

Solution (Exam Style):

Step 1 (Domain): Given,

$$f(x) = \frac{x-2}{3-x}$$

The function is defined for all real x except where the denominator is zero:

$$3 - x = 0$$
$$\implies x = 3$$

$\therefore$ Domain, $D_f = \mathbb{R} \setminus \{3\}$.

Step 2 (Range): Let,

$$y = \frac{x-2}{3-x}$$
$$\implies y(3-x) = x - 2$$
$$\implies 3y - xy = x - 2$$
$$\implies x + xy = 3y + 2 \quad \text{[Grouping all } x \text{ terms on the left]}$$
$$\implies x(1+y) = 3y + 2$$
$$\implies x = \frac{3y+2}{y+1} \quad \text{[Undefined when } y + 1 = 0 \implies y = -1\text{]}$$

$\therefore$ Range, $R_f = \mathbb{R} \setminus \{-1\}$.

Problem 1.4 (Class Homework): Find the domain and range of $f(x) = \sqrt{\dfrac{x-2}{3-x}}$.

Solution (Exam Style):

Step 1 (Domain): For $f(x)$ to be real and finite:

$$\frac{x-2}{3-x} \geq 0 \quad \text{and} \quad 3-x \neq 0 \quad \text{[Critical points: } x=2 \text{ and } x=3\text{]}$$

Sign analysis across the intervals $(-\infty, 2)$, $(2, 3)$, and $(3, \infty)$:

Interval	Sign of $(x-2)$	Sign of $(3-x)$	Sign of $\dfrac{x-2}{3-x}$	Status
$(-\infty, 2)$	$-$	$+$	$-$	Invalid (< 0)
$(2, 3)$	$+$	$+$	$+$	Valid (> 0)
$(3, \infty)$	$+$	$-$	$-$	Invalid (< 0)

Checking endpoints:

At $x = 2$, $\quad f(2) = \sqrt{\dfrac{2-2}{3-2}} = 0$ [Defined, so $x = 2$ is included]

At $x = 3$, $\quad f(3) = \sqrt{\dfrac{3-2}{3-3}} = \sqrt{\dfrac{1}{0}}$ [Undefined, so $x = 3$ is excluded]

$\therefore$ Domain, $D_f = [2, 3)$.

Step 2 (Range): Let $y = \sqrt{\dfrac{x-2}{3-x}}$, where $y \geq 0$ (since it is a principal square root). Squaring both sides:

$$\begin{aligned} y^2 &= \frac{x-2}{3-x} \\ &\implies 3y^2 - xy^2 = x - 2 \\ &\implies x(1+y^2) = 3y^2 + 2 \\ &\implies x = \frac{3y^2+2}{y^2+1} \quad \text{[Defined for all real } y\text{, and } y \geq 0\text{]} \end{aligned}$$

$\therefore$ Range, $R_f = [0, \infty)$.

Problem 1.5 (Class Lecture): Find the domain and range of $f(x) = \dfrac{1}{\sqrt{x^2 - 5x + 6}}$.

Solution (Exam Style): Since the square root is in the denominator, the quantity inside the radical must be strictly positive:

$$\begin{aligned} x^2 - 5x + 6 &> 0 \\ &\implies (x-2)(x-3) > 0 \quad \text{[Both endpoints } x = 2, 3 \text{ are excluded]} \end{aligned}$$

From the sign table in Problem 1.2, $(x-2)(x-3) > 0$ on $(-\infty, 2)$ and $(3, \infty)$, and $f(x) > 0$ everywhere on its domain.

$\therefore$ Domain, $D_f = (-\infty, 2) \cup (3, \infty)$, $\qquad$ Range, $R_f = (0, \infty)$.

Problem 1.6 (PUST EECE 2021–22 Exam Q1a [5 Marks]): Find the domain and range of $f(x) = \sqrt{16 - x^2}$ and describe its graph.

Solution (Exam Style):

For Domain: $\quad 16 - x^2 \geq 0$

$$\begin{aligned} &\implies x^2 \leq 16 \\ &\implies -4 \leq x \leq 4 \end{aligned}$$

$\therefore$ Domain, $D_f = [-4, 4]$.

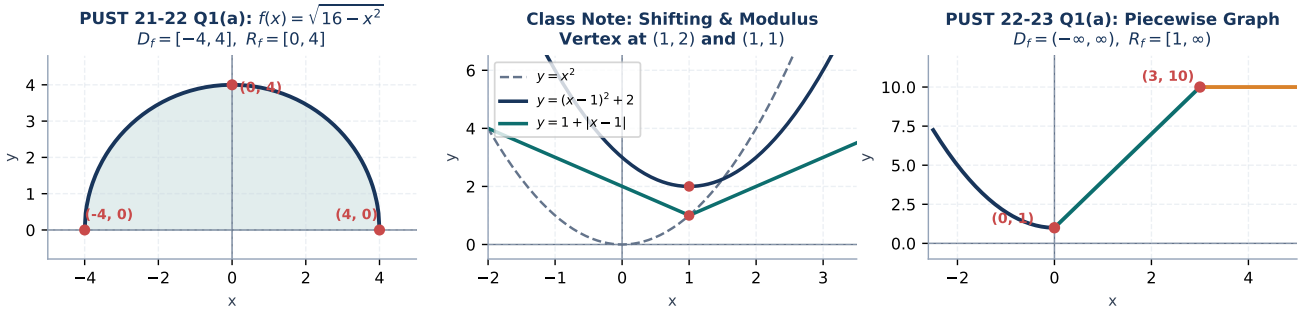
For Range: $0 \le x^2 \le 16$ $[\because x \in [-4, 4]]$

$\implies 0 \le 16 - x^2 \le 16$

$\implies 0 \le \sqrt{16 - x^2} \le 4$

$\therefore$ Range, $R_f = [0, 4]$.

Graph: Setting $y = \sqrt{16 - x^2}$ with $y \ge 0$ gives $x^2 + y^2 = 4^2$ $(y \ge 0)$, which is the upper semicircle centered at $(0, 0)$ with radius $r = 4$.



1.5 Solved Anton Homework Problems on Domain & Range (Section 0.1: Ex 7–14, 19–20, 25–28, 35–38)

1. **Ex. 8:** $f(x) = \dfrac{1}{5x + 7}$.

   $5x + 7 \ne 0$

   $\implies x \ne -\dfrac{7}{5}$

   $\therefore D_f = \left(-\infty, -\dfrac{7}{5}\right) \cup \left(-\dfrac{7}{5}, \infty\right), \qquad R_f = (-\infty, 0) \cup (0, \infty).$

2. **Ex. 9 & 10:** (a) $g(x) = \sqrt{x^2 - 3}$ and (b) $g(x) = \sqrt{x^2 + 3}$.

   (a) $x^2 - 3 \ge 0$

   $\implies |x| \ge \sqrt{3}$

   $\therefore D_g = (-\infty, -\sqrt{3}] \cup [\sqrt{3}, \infty), \qquad R_g = [0, \infty).$

   (b) $x^2 + 3 \ge 3 > 0 \quad \forall x \in \mathbb{R}$

   $\therefore D_g = (-\infty, \infty), \qquad R_g = [\sqrt{3}, \infty).$

3. **Ex. 12:** $h(x) = \sqrt{x - 3x^2}$.

   $x - 3x^2 \ge 0$

   $\implies x(1 - 3x) \ge 0$

   $\therefore D_h = \left[0, \dfrac{1}{3}\right], \qquad R_h = \left[0, \dfrac{1}{2\sqrt{3}}\right]$ [Max at $x = \frac{1}{6}$ is $\sqrt{\frac{1}{12}} = \frac{1}{2\sqrt{3}}$].

4. **Ex. 13:** $\phi(x) = \dfrac{x}{|x| + 1}$.

   $|x| + 1 \ge 1 \ne 0 \quad \forall x \in \mathbb{R}$

   $\therefore D_\phi = (-\infty, \infty)$

$$0 \le \frac{|x|}{|x|+1} < 1$$
$$\implies -1 < \frac{x}{|x|+1} < 1$$
$$\therefore R_\phi = (-1, 1).$$

5. **Ex. 14:** $F(x) = 3 - \sqrt{x-2}$.

$$x - 2 \ge 0$$
$$\implies x \ge 2$$
$$\therefore D_F = [2, \infty)$$
$$\sqrt{x-2} \ge 0$$
$$\implies 3 - \sqrt{x-2} \le 3$$
$$\therefore R_F = (-\infty, 3].$$

6. **Ex. 19:** $f(x) = \sqrt{x^2 - 2x + 5}$.

$$x^2 - 2x + 5 = (x-1)^2 + 4 \ge 4 > 0 \quad \forall x \in \mathbb{R} \quad \text{[Completing the square]}$$
$$\therefore D_f = (-\infty, \infty), \qquad R_f = [\sqrt{4}, \infty) = [2, \infty).$$

7. **Ex. 35:** $f(x) = \dfrac{x^2 - 4}{x - 2}$.

$$x - 2 \ne 0$$
$$\implies x \ne 2$$
$$\therefore D_f = \mathbb{R} \setminus \{2\}$$
$$\text{For } x \ne 2, \quad f(x) = \frac{(x-2)(x+2)}{x-2} = x + 2 \quad \text{[Missing value at } x = 2 \text{ is } y = 4\text{]}$$
$$\therefore R_f = \mathbb{R} \setminus \{4\}.$$

2 Unit 2: Graphs of Functions, Inverse, Even, Odd & Periodic Functions

2.1 Graph Transformations: Shifting, Scaling, and Reflecting

Given the graph of a parent function $y = f(x)$ and a positive constant $c > 0$:

1. **Shifting (Translation):**
   - $y = f(x) + c$: Shifts the graph **upward** by c units (e.g., $y = x^2 + 1$, vertex at $(0, 1)$).
   - $y = f(x) - c$: Shifts the graph **downward** by c units (e.g., $y = x^2 - 1$, vertex at $(0, -1)$).
   - $y = f(x - c)$: Shifts the graph to the **right** by c units (e.g., $y = (x-1)^2$, vertex at $(1, 0)$).
   - $y = f(x + c)$: Shifts the graph to the **left** by c units (e.g., $y = (x+1)^2$, vertex at $(-1, 0)$).
   - $y = (x-1)^2 + 2$: Shifts right by 1 unit and upward by 2 units, placing the vertex at $(1, 2)$.
2. **Scaling (Stretching and Compressing):**
   - $y = c\, f(x)$ $(c > 1)$: Stretches the graph vertically away from the x-axis by a factor of c.
   - $y = f(cx)$ $(c > 1)$: Compresses the graph horizontally toward the y-axis by a factor of $1/c$.
   - $y = f(x/c)$ $(c > 1)$: Stretches the graph horizontally away from the y-axis by a factor of c.
3. **Reflecting:**
   - $y = -f(x)$: Reflects the graph across the x-**axis** (e.g., $y = -\sqrt{x}$ in Quadrant IV).
   - $y = f(-x)$: Reflects the graph across the y-**axis** (e.g., $y = \sqrt{-x}$ in Quadrant II).

Problem 2.1 (Class Lecture): Analyze and sketch $f(x) = 1 + |x - 1|$.

Solution (Exam Style): By definition of absolute value:

$$|x-1| = \begin{cases} -(x-1) = 1-x, & \text{if } x < 1 \\ x-1, & \text{if } x \geq 1 \end{cases}$$

Adding 1 to both branches:

$$f(x) = 1 + |x-1| = \begin{cases} 2-x, & \text{if } x < 1 \\ x, & \text{if } x \geq 1 \end{cases}$$

- **Graph:** Start with $y = |x|$ (V-shape at $(0,0)$), shift 1 unit right to $(1,0)$, and shift 1 unit up to get a V-shaped graph with vertex at $(1,1)$ and y-intercept at $(0,2)$.
- **Domain & Range:** $D_f = (-\infty, \infty)$ and $R_f = [1, \infty)$.

2.2 Piecewise-Defined Functions

PUST EECE 2022–23 Semester Exam — Question 1(a) [4 Marks]

Question: Define a piecewise function. Sketch the graph of the following function and mention the domain and range from the graph:

$$f(x) = \begin{cases} x^2 + 1, & x < 0 \\ 3x + 1, & 0 \leq x \leq 3 \\ 10, & x > 3 \end{cases}$$

Solution (Exam Style):

- **Definition:** A *piecewise-defined function* is a function defined by multiple sub-functions, where each sub-function applies to a distinct interval of the domain.
- **Interval-by-Interval Analysis for Sketching:**
  1. **For $x < 0$:** $y = x^2 + 1$ is the left half of an upward-opening parabola approaching $(0,1)$ as $x \to 0^-$ and rising to $+\infty$ as $x \to -\infty$. Thus $y \in (1, \infty)$.
  2. **For $0 \leq x \leq 3$:** $y = 3x + 1$ is a straight line segment joining $(0,1)$ to $(3,10)$. Thus $y \in [1, 10]$.
  3. **For $x > 3$:** $y = 10$ is a horizontal ray starting at $(3,10)$ and extending to $+\infty$ parallel to the x-axis.
- **Domain and Range from the Graph:**

  Domain, $D_f = (-\infty, 0) \cup [0, 3] \cup (3, \infty) = (-\infty, \infty) = \mathbb{R}$,

  Range, $R_f = [1, \infty)$ [Minimum $y = 1$ at $x = 0$, unbounded above].

2.3 Inverse Functions

Theorem (Existence of Inverse Function): A function $f : A \to B$ possesses an inverse function $f^{-1} : B \to A$ if and only if f is **bijective** (both **One-to-One** and **Onto**):

1. **One-to-One (Injective):**
   - *Horizontal Line Test:* Every horizontal line intersects the graph of $y = f(x)$ at most once.
   - *Algebraic Test:* $f(x_1) = f(x_2) \implies x_1 = x_2$ for all $x_1, x_2 \in D_f$.
2. **Onto (Surjective):** Range(f) = Co-domain(f).

Properties of $f^{-1}(x)$:

- $D_{f^{-1}} = R_f$ and $R_{f^{-1}} = D_f$.
- The graph of $y = f^{-1}(x)$ is the reflection of $y = f(x)$ about the line $y = x$.

Problem 2.2 (Class Lecture & Anton Sec. 0.4 Homework Ex 19–26, 37–38): Find the inverse function $f^{-1}(x)$ for each of the following:

1. **Class Problem:** $f(x) = 3x - 2$.

   Solution:

   One-to-One Check: $f(x_1) = f(x_2)$
   $\implies 3x_1 - 2 = 3x_2 - 2$
   $\implies x_1 = x_2$ [Hence f is one-to-one]

   To find $f^{-1}(x)$, let $y = f(x)$:

   $y = 3x - 2$
   $\implies 3x = y + 2$
   $\implies x = \dfrac{y+2}{3}$
   $\therefore f^{-1}(x) = \dfrac{x+2}{3}$ [Replacing y by x]

2. **Anton 0.4 Ex. 20:** $f(x) = \dfrac{x+1}{x-1}$.

   Solution:

   Let $y = \dfrac{x+1}{x-1}$
   $\implies yx - y = x + 1$
   $\implies x(y-1) = y + 1$
   $\implies x = \dfrac{y+1}{y-1}$
   $\therefore f^{-1}(x) = \dfrac{x+1}{x-1}, \quad x \neq 1.$

3. **Anton 0.4 Ex. 21 & 22:** (a) $f(x) = 3x^3 - 5$ and (b) $f(x) = \sqrt[5]{4x+2}$.

   Solution:

   (a) Let $y = 3x^3 - 5$
   $\implies x^3 = \dfrac{y+5}{3}$
   $\implies x = \sqrt[3]{\dfrac{y+5}{3}}$
   $\therefore f^{-1}(x) = \sqrt[3]{\dfrac{x+5}{3}}.$

   (b) Let $y = \sqrt[5]{4x+2}$
   $\implies y^5 = 4x + 2$
   $\implies x = \dfrac{y^5 - 2}{4}$
   $\therefore f^{-1}(x) = \dfrac{x^5 - 2}{4}.$

4. **Anton 0.4 Ex. 23 & 25 (Domain-Restricted Inverses):** (a) $f(x) = \dfrac{3}{x^2}$ $(x < 0)$ and (b) $f(x) = x^2 - 2x + 1$ $(x \geq 1)$.

   Solution:

   (a) Let $y = \dfrac{3}{x^2}, \quad y > 0$
   $\implies x^2 = \dfrac{3}{y}$

$$\implies x = -\sqrt{\frac{3}{y}} \quad \text{[Negative root chosen since } x < 0\text{]}$$

$$\therefore f^{-1}(x) = -\sqrt{\frac{3}{x}}, \quad x > 0.$$

(b) Let $y = (x-1)^2, \quad y \geq 0$

$$\implies x - 1 = +\sqrt{y} \quad \text{[Positive root chosen since } x \geq 1\text{]}$$

$$\implies x = 1 + \sqrt{y}$$

$$\therefore f^{-1}(x) = 1 + \sqrt{x}, \quad x \geq 0.$$

5. **Anton 0.4 Ex. 37:** $f(x) = \frac{1}{2}(e^x + e^{-x})$, $x \geq 0$.

   Solution:

Let $y = \dfrac{e^x + e^{-x}}{2}$ [For $x \geq 0$, $y \geq 1$]

$$\implies 2y = e^x + \frac{1}{e^x}$$

$$\implies (e^x)^2 - 2y(e^x) + 1 = 0 \quad \text{[Quadratic equation in } e^x\text{]}$$

$$\implies e^x = \frac{2y \pm \sqrt{4y^2 - 4}}{2} = y \pm \sqrt{y^2 - 1}$$

$$\implies e^x = y + \sqrt{y^2 - 1} \quad [\because x \geq 0 \implies e^x \geq 1]$$

$$\implies x = \ln\left(y + \sqrt{y^2 - 1}\right)$$

$$\therefore f^{-1}(x) = \ln\left(x + \sqrt{x^2 - 1}\right), \quad x \geq 1.$$

(Similarly, for Ex. 38, $f(x) = \frac{1}{2}(e^x - e^{-x})$ gives $(e^x)^2 - 2y(e^x) - 1 = 0 \implies e^x = y + \sqrt{y^2+1} \implies f^{-1}(x) = \ln(x + \sqrt{x^2+1})$).

2.4 Even, Odd, and Periodic Functions (PUST EECE 2020–21 Q1a)

1. **Definition (Even Function):** A function $f(x)$ is called an **even function** if for all $x \in D_f$:

   $$f(-x) = f(x)$$

   - *Symmetry:* The graph of an even function is symmetric about the y-**axis**.
   - *Examples:* $f(x) = x^2$, $f(x) = \cos x$, $f(x) = |x|$, $f(x) = x^4 - 3x^2 + 1$.

2. **Definition (Odd Function):** A function $f(x)$ is called an **odd function** if for all $x \in D_f$:

   $$f(-x) = -f(x)$$

   - *Symmetry:* The graph of an odd function is symmetric about the **origin** $(0, 0)$.
   - *Examples:* $f(x) = x^3$, $f(x) = \sin x$, $f(x) = \tan x$, $f(x) = x^5 - x$.

3. **Definition (Periodic Function):** A function $f(x)$ is called **periodic** if there exists a positive real constant $T > 0$ such that:

   $$f(x + T) = f(x) \quad \forall x \in D_f.$$

   The smallest such positive number T is called the **fundamental period** of $f(x)$.

   - Period of $\sin(ax+b), \cos(ax+b), \sec(ax+b), \csc(ax+b)$ is $T = \dfrac{2\pi}{|a|}$.
   - Period of $\tan(ax+b), \cot(ax+b)$ is $T = \dfrac{\pi}{|a|}$.

3 Unit 3: Limits and Continuity of Functions

3.1 Concept of Limit and Existence Criterion

Definition (Limit): If $f(x)$ approaches a single finite real number L as x approaches a from both the left ($x < a$) and the right ($x > a$) with $x \neq a$, then L is called the limit of $f(x)$ as $x \to a$:

$$\lim_{x \to a} f(x) = L.$$

Theorem (Existence of Limit): The two-sided limit $\lim_{x \to a} f(x) = L$ exists if and only if both the Left-Hand Limit (L.H.L.) and Right-Hand Limit (R.H.L.) exist as finite numbers and are equal:

$$\text{L.H.L.} = \lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = \text{R.H.L.} = L.$$

Problem 3.1 (Class Lecture): Examine the existence of limits at $x = 0$ and $x = 3$ for:

$$f(x) = \begin{cases} x^2 + 5, & x < 0 \\ x + 5, & 0 \leq x \leq 3 \\ 2x + 1, & x > 3 \end{cases}$$

Solution (Exam Style):

(i) At $x = 0$:

$$\text{L.H.L.} = \lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (x^2 + 5) = 0^2 + 5 = 5 \quad [\text{For } x < 0,\ f(x) = x^2 + 5]$$
$$\text{R.H.L.} = \lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} (x + 5) = 0 + 5 = 5 \quad [\text{For } 0 \leq x \leq 3,\ f(x) = x + 5]$$

Since L.H.L. = R.H.L. = 5, $\lim_{x \to 0} f(x) = 5$ exists.

(ii) At $x = 3$:

$$\text{L.H.L.} = \lim_{x \to 3^-} f(x) = \lim_{x \to 3^-} (x + 5) = 3 + 5 = 8 \quad [\text{For } x \leq 3,\ f(x) = x + 5]$$
$$\text{R.H.L.} = \lim_{x \to 3^+} f(x) = \lim_{x \to 3^+} (2x + 1) = 2(3) + 1 = 7 \quad [\text{For } x > 3,\ f(x) = 2x + 1]$$

Since L.H.L. $\neq$ R.H.L. ($8 \neq 7$), $\lim_{x \to 3} f(x)$ **does not exist**.

Problem 3.2 (Class Lecture): Evaluate $\lim_{x \to 3} \dfrac{|x - 3|}{x - 3}$.

Solution (Exam Style):

$$\begin{aligned} \text{L.H.L.} &= \lim_{x \to 3^-} \frac{|x - 3|}{x - 3} \\ &= \lim_{x \to 3^-} \frac{-(x - 3)}{x - 3} \quad [\because \text{for } x < 3,\ |x - 3| = -(x - 3)] \\ &= -1 \end{aligned}$$

$$\begin{aligned} \text{R.H.L.} &= \lim_{x \to 3^+} \frac{|x - 3|}{x - 3} \\ &= \lim_{x \to 3^+} \frac{+(x - 3)}{x - 3} \quad [\because \text{for } x > 3,\ |x - 3| = +(x - 3)] \\ &= +1 \end{aligned}$$

Since L.H.L. $\neq$ R.H.L., $\lim_{x \to 3} \dfrac{|x - 3|}{x - 3}$ **does not exist**.

3.2 Formal (ϵ-δ) Definition of Limit

Definition (ϵ-δ Definition): Let $f(x)$ be defined on an open interval containing a, except possibly at $x = a$. Then:

$$\lim_{x \to a} f(x) = L$$

if for every given $\epsilon > 0$, there exists a $\delta > 0$ such that:

$$|f(x) - L| < \epsilon \qquad \text{whenever} \qquad 0 < |x - a| < \delta.$$

Class Lecture & PUST EECE 2020–21 Exam Q1(c) [4.5 Marks]
Question: Using the ϵ-δ definition of limit, prove that $\lim_{x \to 2}(3x - 5) = 1$.

Proof (Exam Style): Let $\epsilon > 0$ be any arbitrary positive real number. Here, $f(x) = 3x - 5$, $a = 2$, and $L = 1$. We must find a $\delta > 0$ such that:

$$|(3x - 5) - 1| < \epsilon \qquad \text{whenever} \qquad 0 < |x - 2| < \delta.$$

Simplifying the left inequality vertically:

$$\begin{aligned} &|(3x - 5) - 1| < \epsilon \\ &\implies |3x - 6| < \epsilon \\ &\implies |3(x - 2)| < \epsilon \\ &\implies 3|x - 2| < \epsilon \\ &\implies |x - 2| < \frac{\epsilon}{3} \end{aligned}$$

Choose $\delta = \dfrac{\epsilon}{3} > 0$. Then whenever $0 < |x - 2| < \delta$:

$$\begin{aligned} |(3x - 5) - 1| &= |3x - 6| \\ &= 3|x - 2| \\ &< 3\delta \quad [\because |x - 2| < \delta] \\ &= 3\left(\frac{\epsilon}{3}\right) = \epsilon. \end{aligned}$$

$\therefore$ By the ϵ-δ definition of limit, $\lim_{x \to 2}(3x - 5) = 1$. ■

3.3 Limits at Infinity, Infinite Limits & Trigonometric Limits

Standard Limit Formulas:

$$\lim_{x \to \pm\infty} \frac{1}{x^n} = 0 \ (n > 0), \qquad \lim_{x \to 0} \frac{\sin x}{x} = 1, \qquad \lim_{x \to 0} \frac{\tan x}{x} = 1, \qquad \lim_{x \to 0} \frac{1 - \cos x}{x} = 0.$$

Theorem (Squeezing / Sandwich Theorem): If $g(x) \le f(x) \le h(x)$ for all $x \neq a$ in an open interval containing a, and $\lim_{x \to a} g(x) = \lim_{x \to a} h(x) = L$, then $\lim_{x \to a} f(x) = L$.

Problem 3.3 (Class Lecture): Evaluate $\lim_{x \to \infty} \dfrac{(2x + 3)^3 (3x - 2)^2}{x^5 + 5}$.

Solution (Exam Style):

$$\begin{aligned} L &= \lim_{x \to \infty} \frac{(2x + 3)^3 (3x - 2)^2}{x^5 + 5} \\ &= \lim_{x \to \infty} \frac{\dfrac{(2x + 3)^3}{x^3} \cdot \dfrac{(3x - 2)^2}{x^2}}{\dfrac{x^5 + 5}{x^5}} \quad \text{[Dividing numerator and denominator by } x^5 = x^3 \cdot x^2\text{]} \end{aligned}$$

$$= \lim_{x\to\infty} \frac{\left(2+\frac{3}{x}\right)^3\left(3-\frac{2}{x}\right)^2}{1+\frac{5}{x^5}}$$
$$= \frac{(2+0)^3(3-0)^2}{1+0} \quad [\because \text{as } x\to\infty,\ \tfrac{1}{x}\to 0 \text{ and } \tfrac{1}{x^5}\to 0]$$
$$= 8\times 9 = 72.$$

3.4 Solved Anton Homework Sets 1.1, 1.2, 1.3 & 1.6 (Vertical Step-by-Step Solutions)

A. Exercise Set 1.1 (Problems 1, 2, 9 — Graphical Limits)

- **Ex. 1.1 #1:** From the given graph of $g(x)$:

  (a) $\lim_{x\to 0^-} g(x) = -1$, (b) $\lim_{x\to 0^+} g(x) = -2$,
  (c) $\lim_{x\to 0} g(x)$ does not exist ($\because$ L.H.L. $\neq$ R.H.L.), (d) $g(0) = 2$.

- **Ex. 1.1 #2:** From the given graph of $G(x)$:

  (a) $\lim_{x\to 0^-} G(x) = 1$, (b) $\lim_{x\to 0^+} G(x) = 1$,
  (c) $\lim_{x\to 0} G(x) = 1$ ($\because$ L.H.L. = R.H.L.), (d) $G(0) = 3$.

- **Ex. 1.1 #9:** From the given graph of $f(x)$:

  (a) $\lim_{x\to -2} f(x) = +\infty$, (b) $\lim_{x\to 0^-} f(x) = 2$, (c) $\lim_{x\to 0^+} f(x) = +\infty$,
  (d) $\lim_{x\to 2^-} f(x) = -\infty$, (e) $\lim_{x\to 2^+} f(x) = 0$, (f) Vertical Asymptotes: $x=-2$, $x=0$, $x=2$.

B. Exercise Set 1.2 (Problems 3–32 — Algebraic, Infinite & Radical Limits)

1. **Ex. 7, 8, 9, 11, 12 (Factoring $0/0$ Polynomial Quotients):**

   Ex. 8: $\lim_{t\to -2} \frac{t^3+8}{t+2} = \lim_{t\to -2} \frac{(t+2)(t^2-2t+4)}{t+2}$ $[a^3+b^3=(a+b)(a^2-ab+b^2)]$
   $= \lim_{t\to -2}(t^2-2t+4)$ $[\because t\neq -2]$
   $= (-2)^2 - 2(-2) + 4 = 12.$

   Ex. 12: $\lim_{x\to 1} \frac{3x^2-x-2}{2x^2+x-3} = \lim_{x\to 1} \frac{(x-1)(3x+2)}{(x-1)(2x+3)}$
   $= \lim_{x\to 1} \frac{3x+2}{2x+3} = \frac{3(1)+2}{2(1)+3} = 1.$

   *(By the same vertical factoring method: **Ex. 3:** 6; **Ex. 4:** 27; **Ex. 5:** $\frac{3}{4}$; **Ex. 6:** -3; **Ex. 7:** 4; **Ex. 9:** $-\frac{4}{5}$; **Ex. 10:** 0; **Ex. 11:** -3.)*

2. **Ex. 13 & 14 (Cubic Factorization):**

   Ex. 13: $\lim_{t\to 2} \frac{t^3+3t^2-12t+4}{t^3-4t} = \lim_{t\to 2} \frac{(t-2)(t^2+5t-2)}{t(t-2)(t+2)}$ [Factoring $(t-2)$]
   $= \lim_{t\to 2} \frac{t^2+5t-2}{t(t+2)} = \frac{4+10-2}{2(4)} = \frac{12}{8} = \frac{3}{2}.$

   Ex. 14: $\lim_{t\to 1} \frac{t^3+t^2-5t+3}{t^3-3t+2} = \lim_{t\to 1} \frac{(t-1)^2(t+3)}{(t-1)^2(t+2)}$ [Double root at $t=1$]
   $= \lim_{t\to 1} \frac{t+3}{t+2} = \frac{4}{3}.$

3. **Ex. 15–28 (One-Sided Infinite Limits):**

Ex. 24–26: $\lim_{x\to 4^+} \dfrac{3-x}{x^2-2x-8} = \lim_{x\to 4^+} \dfrac{3-x}{(x-4)(x+2)}$

$= -\infty$ [Numerator $\to -1 < 0$, denominator $\to 0^+$]

$\lim_{x\to 4^-} \dfrac{3-x}{(x-4)(x+2)} = +\infty$ [Numerator $\to -1 < 0$, denominator $\to 0^-$]

$\therefore \lim_{x\to 4} \dfrac{3-x}{x^2-2x-8}$ does not exist.

*(Similarly: **Ex. 15:** $+\infty$; **Ex. 16:** $-\infty$; **Ex. 17:** DNE; **Ex. 18:** $+\infty$; **Ex. 19:** $-\infty$; **Ex. 20:** DNE; **Ex. 21:** $+\infty$; **Ex. 22:** $-\infty$; **Ex. 23:** DNE; **Ex. 27:** $+\infty$; **Ex. 28:** $+\infty$.)*

4. **Ex. 29 & 30 (Rationalizing Radicals):**

Ex. 29: $\lim_{x\to 9} \dfrac{x-9}{\sqrt{x}-3} = \lim_{x\to 9} \dfrac{(\sqrt{x}-3)(\sqrt{x}+3)}{\sqrt{x}-3}$ [$\because x-9 = (\sqrt{x})^2 - 3^2$]

$= \lim_{x\to 9}(\sqrt{x}+3) = \sqrt{9}+3 = 6.$

Ex. 30: $\lim_{y\to 4} \dfrac{4-y}{2-\sqrt{y}} = \lim_{y\to 4}(2+\sqrt{y}) = 2+2 = 4.$

5. **Ex. 31 & 32 (Piecewise Limits):**

Ex. 31: $\lim_{x\to 3^-}(x-1) = 2, \quad \lim_{x\to 3^+}(3x-7) = 2 \implies \lim_{x\to 3} f(x) = 2.$

Ex. 32: (a) At $t=0$: LHL $= -2 \neq$ RHL $= 0 \implies$ DNE; (b) $\lim_{t\to 1} g(t) = 1$; (c) $\lim_{t\to 2} g(t) = 4.$

C. Exercise Set 1.3 (Problems 9–32, 39, 40 — Limits at $\pm\infty$)

Important Note: Square Roots as $x \to -\infty$

When factoring x^2 out of a square root, always write $\sqrt{x^2} = |x| = \begin{cases} +x, & \text{if } x > 0 \ (x \to +\infty) \\ -x, & \text{if } x < 0 \ (x \to -\infty) \end{cases}$.

1. **Ex. 25 & 26:** Evaluate $\lim_{x\to-\infty} \dfrac{\sqrt{5x^2-2}}{x+3}$ and $\lim_{x\to+\infty} \dfrac{\sqrt{5x^2-2}}{x+3}$.

Ex. 25 ($x \to -\infty$): $L = \lim_{x\to-\infty} \dfrac{\sqrt{x^2\left(5-\frac{2}{x^2}\right)}}{x\left(1+\frac{3}{x}\right)}$

$= \lim_{x\to-\infty} \dfrac{|x|\sqrt{5-\frac{2}{x^2}}}{x\left(1+\frac{3}{x}\right)}$

$= \lim_{x\to-\infty} \dfrac{-x\sqrt{5-\frac{2}{x^2}}}{x\left(1+\frac{3}{x}\right)}$ [$\because$ for $x<0$, $|x| = -x$]

$= -\dfrac{\sqrt{5-0}}{1+0} = -\sqrt{5}.$

Ex. 26 ($x \to +\infty$): $L = \lim_{x\to+\infty} \dfrac{+x\sqrt{5-\frac{2}{x^2}}}{x\left(1+\frac{3}{x}\right)} = +\sqrt{5}$ [$\because$ for $x>0$, $|x| = +x$].

*(Similarly: **Ex. 27:** $+\frac{1}{\sqrt{6}}$; **Ex. 28:** $-\frac{1}{\sqrt{6}}$; **Ex. 29 & 30:** $+\sqrt{3}$ since $\sqrt{x^4} = x^2 > 0$.)*

2. **Ex. 31 & 32 (Indeterminate $\infty - \infty$ via Conjugate):**

Ex. 32: $L = \lim_{x\to+\infty} \left(\sqrt{x^2-3x} - x\right)$

$$= \lim_{x\to+\infty} \frac{\left(\sqrt{x^2-3x}-x\right)\left(\sqrt{x^2-3x}+x\right)}{\sqrt{x^2-3x}+x} \quad \text{[Multiplying by conjugate]}$$
$$= \lim_{x\to+\infty} \frac{(x^2-3x)-x^2}{x\left(\sqrt{1-\frac{3}{x}}+1\right)}$$
$$= \lim_{x\to+\infty} \frac{-3}{\sqrt{1-\frac{3}{x}}+1} = -\frac{3}{1+1} = -\frac{3}{2}.$$

*(For **Ex. 31**, $\lim_{x\to+\infty}(\sqrt{x^2+3}-x) = \lim_{x\to+\infty}\frac{3}{\sqrt{x^2+3}+x} = 0$. Other Set 1.3 answers: **9.** $-\infty$; **10.** $+\infty$; **11.** $+\infty$; **12.** $+\infty$; **13.** $\frac{3}{2}$; **14.** $\frac{5}{2}$; **15–17.** 0; **18.** $\frac{5}{3}$; **19.** $-\infty$; **20.** $+\infty$; **21.** $-\frac{1}{7}$; **22.** $\frac{4}{7}$; **23.** $-\frac{\sqrt[3]{5}}{2}$; **24.** $\sqrt[3]{\frac{3}{2}}$; **39(a)** $+\infty$, **(b)** -5; **40(a)** 0, **(b)** -6.)*

D. Exercise Set 1.6 (Problems 1–8, 11–32, 41, 42 — Trigonometric Limits & Continuity)

1. **Ex. 19, 27, 28, 30 (Representative Trigonometric Limit Derivations):**

Ex. 19: $\lim_{x\to0}\frac{\tan 7x}{\sin 3x} = \lim_{x\to0}\left(\frac{\tan 7x}{7x}\cdot\frac{3x}{\sin 3x}\cdot\frac{7}{3}\right) = 1\cdot1\cdot\frac{7}{3} = \frac{7}{3}.$

Ex. 27: $\lim_{\theta\to0}\frac{\theta^2}{1-\cos\theta} = \lim_{\theta\to0}\frac{\theta^2(1+\cos\theta)}{1-\cos^2\theta}$ [Multiplying by $1+\cos\theta$]

$= \lim_{\theta\to0}\left(\frac{\theta}{\sin\theta}\right)^2(1+\cos\theta) = (1)^2(1+1) = 2.$

Ex. 28: $\lim_{h\to0}\frac{1-\cos 3h}{\cos^2 5h-1} = \lim_{h\to0}\frac{2\sin^2(3h/2)}{-\sin^2 5h}$ $[1-\cos 3h = 2\sin^2(3h/2)]$

$= -2\lim_{h\to0}\frac{\left(\frac{\sin(3h/2)}{3h/2}\right)^2\cdot\frac{9h^2}{4}}{\left(\frac{\sin 5h}{5h}\right)^2\cdot 25h^2} = -2\cdot\frac{9/4}{25} = -\frac{9}{50}.$

*(Other Set 1.6 answers: **1.** none; **2.** $x=\pi$; **3.** $x=n\pi$; **4.** $x=\frac{\pi}{2}+n\pi$; **5.** $x=n\pi$; **6.** none; **7.** $x=\frac{\pi}{6}+2n\pi, \frac{5\pi}{6}+2n\pi$; **8.** $x=\frac{\pi}{2}+n\pi$; **11.** 1; **12.** $-\frac{\sqrt{3}}{2}$; **13.** 3; **14.** $\frac{1}{2}$; **15.** -3; **16.** 0; **17.** $+\infty$; **18.** 0; **20.** $\frac{3}{4}$; **21.** 0; **22.** $\frac{1}{3}$; **23.** 0; **24.** DNE; **25.** 1; **26.** 1; **29.** DNE; **30.** 28; **31.** $\frac{a}{b}$; **32.** 0.)*

2. **Ex. 41:** Find $k\neq0$ so that $f(x) = \begin{cases}\frac{\tan kx}{x}, & x<0\\ 3x+2k^2, & x\geq0\end{cases}$ is continuous at $x=0$.

L.H.L. $= \lim_{x\to0^-}\frac{\tan kx}{x} = \lim_{x\to0^-} k\cdot\frac{\tan kx}{kx} = k(1) = k$
R.H.L. $= f(0) = 3(0)+2k^2 = 2k^2$
For continuity at $x=0$: $\quad 2k^2 = k$
$\implies k(2k-1) = 0$
$\implies k = \frac{1}{2}$ $[\because k\neq0]$.

3.5 Continuity and Types of Discontinuity

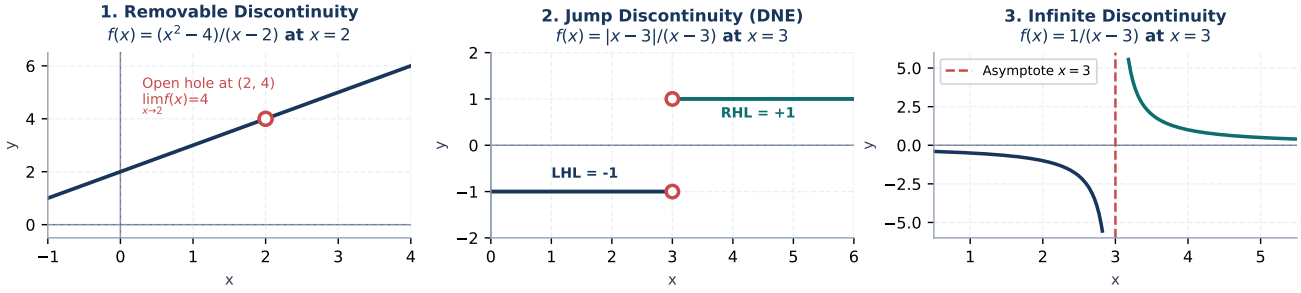
Definition (Continuity at a Point): A function $f(x)$ is continuous at $x=a$ if three conditions hold:

1. $f(a)$ is defined,
2. $\lim_{x\to a} f(x)$ exists (L.H.L. = R.H.L.), and
3. $\lim_{x\to a} f(x) = f(a)$.

Types of Discontinuity:

- **Removable Discontinuity:** $\lim_{x\to a} f(x) = L$ exists, but $f(a)$ is either undefined or $f(a)\neq L$ (removed by setting $f(a)=L$, e.g., $f(x)=\frac{x^2-4}{x-2}$ at $x=2$, where $L=4$).

- **Jump Discontinuity:** L.H.L. and R.H.L. both exist finitely, but L.H.L. $\neq$ R.H.L.
- **Infinite / Essential Discontinuity:** One or both one-sided limits are $\pm\infty$ or do not exist.



PUST EECE 2022–23 Semester Exam — Question 1(b) [4.5 Marks]

Question: What are the conditions for a function to be continuous at $x = a$? For what values of a and b is

$$f(x) = \begin{cases} -2, & x \leq -1 \\ ax - b, & -1 < x < 1 \\ 3, & x \geq 1 \end{cases}$$

continuous at every x?

Solution (Exam Style): Since each piece is polynomial, $f(x)$ is continuous everywhere if it is continuous at $x = -1$ and $x = 1$.

Step 1 (Continuity at $x = -1$):

$$\text{L.H.L.} = \lim_{x \to -1^-} (-2) = -2 = f(-1)$$

$$\text{R.H.L.} = \lim_{x \to -1^+} (ax - b) = -a - b$$

$$\text{Since L.H.L.} = \text{R.H.L.:} \quad -a - b = -2 \implies a + b = 2 \tag{1}$$

Step 2 (Continuity at $x = 1$):

$$\text{L.H.L.} = \lim_{x \to 1^-} (ax - b) = a - b$$

$$\text{R.H.L.} = \lim_{x \to 1^+} (3) = 3 = f(1)$$

$$\text{Since L.H.L.} = \text{R.H.L.:} \quad a - b = 3 \tag{2}$$

Step 3 (Solving for a and b): Adding (1) and (2):

$$2a = 5 \implies a = \frac{5}{2}$$

$$\implies b = 2 - \frac{5}{2} = -\frac{1}{2} \quad \text{[From equation (1)]}$$

$$\therefore a = \frac{5}{2}, \qquad b = -\frac{1}{2}.$$

PUST EECE 2020–21 Semester Exam — Question 1(b) [4.5 Marks]

Question: Find the values of a and b so that $f(x) = \begin{cases} x^2 + 2, & x < 0 \\ ax + b, & 0 \leq x < 1 \\ 3 + 2x - x^2, & x \geq 1 \end{cases}$ is continuous at $x = 0$ and $x = 1$.

Solution (Exam Style):

$$\text{At } x = 0: \quad \text{L.H.L.} = \lim_{x \to 0^-} (x^2 + 2) = 2, \qquad \text{R.H.L.} = f(0) = \lim_{x \to 0^+} (ax + b) = b$$

$$\implies b = 2.$$

At $x = 1$: L.H.L. $= \lim_{x\to 1^-}(ax+b) = a+b$, R.H.L. $= f(1) = \lim_{x\to 1^+}(3+2x-x^2) = 4$

$$\implies a+b = 4$$
$$\implies a+2 = 4 \implies a = 2 \quad [\because b = 2].$$
$$\therefore a = 2, \qquad b = 2.$$

Solved Anton Homework Set 1.5 (Problems 11–22, 29–31, 35, 36, 47)

1. **Ex. 11–22 (Points of Discontinuity): 11–13, 16, 19, 21:** Continuous everywhere (**none**). **14:** $x = \pm 2$. **15:** $x = 0, -\frac{1}{2}$. **17:** $x = 0, \pm 1$. **18:** $x = 0, -4$. **20:** $x^4 + x = x(x+1)(x^2-x+1) = 0 \implies x = 0, -1$. **22:** $x = 1$.

2. **Ex. 29–31 (Constants for Continuity):**

   Ex. 29(b): At $x = 2$, $k(2)^2 = 2(2) + k$

   $$\implies 3k = 4 \implies k = \frac{4}{3}.$$

   Ex. 31: At $x = -1$, $2(-1)^3 - 1 + 7 = m(0) + k \implies k = 4.$

   At $x = 2$, $3m + k = 2^2 + 5$

   $$\implies 3m + 4 = 9 \implies m = \frac{5}{3}.$$

   *(Also: **Ex. 29(a):** $k = 5$; **Ex. 30(a):** $k = 0$; **Ex. 30(b):** no value of k exists.)*

3. **Ex. 35 & 36 (Removable vs. Non-Removable): 35(a)** $x = 0$ (jump, non-removable); **35(b)** $x = -3$ (removable, limit $= -3$); **35(c)** $x = 2$ (removable, limit $= 1$) and $x = -2$ (infinite, non-removable); **36(a)** $x = 2$ (removable, limit $= \frac{1}{3}$); **36(b)** $x = 2$ (jump, non-removable); **36(c)** $x = 1$ (removable, limit $= 8$).

4. **Ex. 47 (Intermediate Value Theorem):** Show $x^3 + x^2 - 2x = 1$ has a root in $[-1, 1]$.

   Let $f(x) = x^3 + x^2 - 2x - 1$ [Continuous on $[-1,1]$]
   $f(-1) = (-1)^3 + (-1)^2 - 2(-1) - 1 = +1 > 0$
   $f(1) = (1)^3 + (1)^2 - 2(1) - 1 = -1 < 0$

   Since $f(1) < 0 < f(-1)$, by the Intermediate Value Theorem there exists $c \in (-1, 1)$ such that $f(c) = 0$. ■

3.6 Past-Question Map for Module 1

Session	Questions Asked from Module 1 (Units 1–3)	Marks
2022–23	Q1(a) Define piecewise function, sketch graph & find domain/range · Q1(b) Continuity conditions & finding a, b	$4 + 4.5$
2021–22	Q1(a) Define function, domain & range; find domain, range & graph of $f(x) = \sqrt{16 - x^2}$	5
2020–21	Q1(a) Even/Odd functions & domain/range of $\frac{x-2}{3-x}$ · Q1(b) Continuity (a, b) · Q1(c) ϵ-δ proof of $\lim_{x\to 2}(3x-5) = 1$	$5 + 4.5 + 4.5$

Quick Revision — Module 1 (Units 1–3)

- **Domain Rules:** $\frac{P}{Q} \implies Q \neq 0$; $\sqrt{g} \implies g \geq 0$; $\frac{1}{\sqrt{g}} \implies g > 0$; $\ln(g) \implies g > 0$.
- **Inverse & Symmetry:** $D_{f^{-1}} = R_f$, $R_{f^{-1}} = D_f$; Even: $f(-x) = f(x)$ (y-axis); Odd: $f(-x) = -f(x)$ (origin).
- **Limits & Continuity:** $\lim_{x\to a} f(x) = L \iff$ LHL $=$ RHL $= L$; ϵ-δ: $0 < |x - a| < \delta \implies |f(x) - L| < \epsilon$; Continuous at $a \iff$ LHL $=$ RHL $= f(a)$.

Common Mistakes to Avoid in the Exam

- Forgetting that $\sqrt{x^2} = |x| = -x$ when $x \to -\infty$ (e.g., Anton 1.3 Ex. 25 gives $-\sqrt{5}$, whereas $x \to +\infty$ gives $+\sqrt{5}$).
- Including denominator zeros in the domain of $\sqrt{\frac{x-2}{3-x}}$: $x = 2$ is included ($[2, 3)$), but $x = 3$ must be strictly excluded!